

$$Ab_3 = A(b_1 + b_2) = Ab_1 + Ab_2 \text{ (Theorem 5(a) from chapter 1)}$$

$\therefore$ The third column of AB is the sum of the first 2 columns of AB . $\square$

20) Suppose the second column of B is all zeros. What can you say about the second column of AB ?

Solution: The second column of B is the zero vector. By the defn. of matrix multiplication, the second column of AB is Ab_2 .

$$Ab_2 = A(0 \cdot b_2) = 0(Ab_2) = 0 \text{ (From Theorem 5(b) of chapter 1)}$$

$\therefore$ The second column of AB is a zero vector.

21) Suppose the last column of AB is entirely zero, but B itself has no column of zeros. What can you say about the columns of A ?

Solution: Let A be a $m \times n$ matrix. Let B have p columns. By defn. of matrix multiplication, B has n rows. AB must be a $m \times p$ matrix.

By defn. of matrix multiplication, ~~Ab_p~~ the p th column of AB is Ab_p .

Given, $Ab_p = 0$. b_p is not a zero column. $\therefore$ The equation $Ab_p = 0$ has a non-trivial solution. $\therefore$ The columns of A are linearly dependent. $\square$

22) Show that if the columns of B are linearly dependent, then so are the columns of AB .

Solution: Let A be a $m \times n$ matrix. Let B have p columns. By defn. of matrix multiplication, B has n rows. AB must be a $m \times p$ matrix.

Let $B = [b_1 \ b_2 \ \dots \ b_p]$. By defn. of linear dependence, there exists $c_1, c_2, \dots, c_p \in \mathbb{R}$ where not all are 0 s.t. $c_1 b_1 + c_2 b_2 + \dots + c_p b_p = 0$.

By the defn. of matrix multiplication, $AB = [Ab_1 \ Ab_2 \ \dots \ Ab_p]$.

$$A(c_1 b_1 + c_2 b_2 + \dots + c_p b_p) = A \cdot 0 \Rightarrow A(c_1 b_1) + A(c_2 b_2) + \dots + A(c_p b_p)$$

$$= A \cdot 0 \text{ (By Repeated Application of Theorem 5(a) on chapter 1)}$$

$$\Rightarrow c_1(Ab_1) + c_2(Ab_2) + \dots + c_p(Ab_p) = 0 \text{ (By Repeated Application of Theorem 5(b) on chapter 2).}$$